

- 6 (a) The table below shows the amount of flour, sugar and number of eggs needed for each type of pastry sold in a cafe.

	Flour (g)	Sugar (g)	Number of eggs
Cookie	90	50	1.5
Cake	220	200	8
Pancake	60	80	2

The above information is represented by a matrix $\mathbf{D} = \begin{pmatrix} 90 & 50 & 1.5 \\ 220 & 200 & 8 \\ 60 & 80 & 2 \end{pmatrix}$.

Each kg of flour and sugar costs \$1.50 and \$1.80 respectively. A dozen eggs costs \$2.70. Complete the cost in \$ for every gram of flour and sugar as well as the cost in \$ for each egg in the table below :

Flour(\$/g)	Sugar(\$/g)	1 Egg(\$)

[2]

- (b) Write down a 3×1 matrix $\mathbf{E}$ such that its elements represent the unit cost of each ingredient needed for the various pastries.

Answer [1]

- (c) Calculate **DE** and state what the elements of **DE** represent.

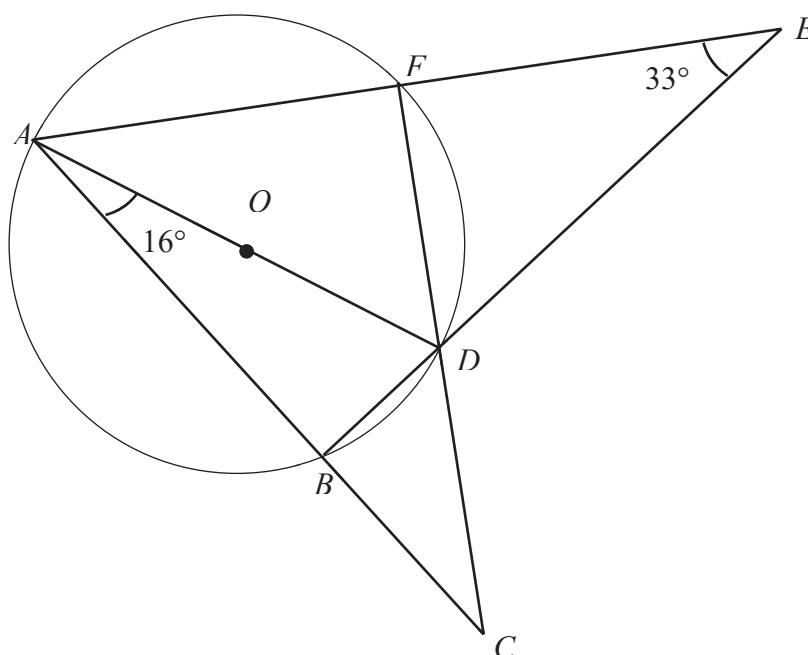
Answer

 [2]

- (d) The cafe prepared 70 cookies, 30 cakes and 120 pancakes on a particular day. Using matrix multiplication, calculate the total cost of the basic ingredients for all the pastries prepared on this day.

Answer \$..... [2]

- 7 (a) In the diagram, AD is the diameter of the circle $ABDF$ with centre O . Given BD and AF produced meet at E , AB and FD produced meet at C , $\angle FED = 33^\circ$ and $\angle DAB = 16^\circ$.



Calculate, stating your reasons clearly,

- (i) $\angle ADB$,

Answer [2]

- (ii) $\angle AFB$,

Answer [1]

- (iii) $\angle DAF$.

Answer [2]